Bank Marketing Campaign Analytics

Marketing portfolio project | Campaign KPI, segmentation, model validation

Business Question

Can historical campaign data support reliable, cost-aware prioritization of future outreach?

Dataset and Scope

UCI Bank Marketing dataset with 41,188 campaign records and 4,640 subscriptions. Records are not verified unique clients.

Method

Built campaign KPIs and a pre-outreach propensity model. Evaluated ranking on the latest 25% of date-ordered records; excluded current-campaign touches and call duration.

Key Findings

Historical conversion rate: 11.3%. Future holdout AUC: 0.53. Top holdout propensity decile: 57.0% conversion versus 25.8% holdout baseline.

Recommendation

Do not deploy this score: chronological AUC is near chance and response by decile is inconsistent. Rebuild with newer data, validate stability, then test a revised rule against a randomized control.